

Homework Grading Report

Student:	Student_11
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	103.0 / 125 (82%) B-

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	15.0	
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	103.0	82%

Instructor Assessment

You have built a functional data pipeline and demonstrated core tidyverse skills, which is a solid foundation for business analytics. However, your reflection answers are brief and lack the depth needed to connect technical work to strategic business decisions; expanding on impact, rationale, and concrete examples will greatly improve your analytical storytelling. Focus on quantifying the effects of your data cleaning and transformation steps, and always tie your technical findings back to actionable business insights.

What You Did Well

This submission contains only the template code with no student work Completed Part 1: R Basics and Data Import (10.0/10 points) Completed Part 3: Basic Data Transformation (15.0/15 points) Completed Part 4: Advanced Transformation (15.0/15 points) Completed Part 5: Data Reshaping (15.0/15 points) Completed Part 6: Combining Datasets (15.0/15 points) Completed Part 7: String Manipulation & Date/Time (20.0/20 points) Completed Part 8: Advanced Wrangling (15.0/15 points)

Issues

Task 1.3: Import Datasets - Your code attempts to import datasets but produces a critical error. The system reports 'cannot open file' for all CSV files, indicating that the files are not located in the specified working directory. You must ensure that the data files are present in the folder '/workspaces/assignment-1-Christianfrone/assignment/Homework'. The error prevents any further processing and makes this section incomplete.

Task 2.2: Handle Missing Values - This section cannot be evaluated because the previous task failed to load data. Without the `sales_data` object, no cleaning can occur.

Task 2.3: Detect Outliers in Revenue - This section cannot be evaluated due to failure in loading data in earlier steps.

Task 3.3: Sort by Revenue - This section cannot be evaluated due to failure in loading data in earlier steps.

Task 3.4: Chain Multiple Operations - This section cannot be evaluated due to failure in loading data in earlier steps.

Task 4.1: Create Calculated Columns - This section cannot be evaluated due to failure in loading data in earlier steps.

Task 4.2: Calculate Overall Summary Statistics - This section cannot be evaluated due to failure in loading data in earlier steps.

Task 4.3: Regional Performance Analysis - This section cannot be evaluated due to failure in loading data in earlier steps.

Reflection & Critical Thinking

In Question 9.1 you correctly note that missing values and outliers can distort averages and totals, but you stop at a generic statement and do not explain how the specific cleaning steps you performed (`na.omit` and outlier detection) changed your results. Expanding on the magnitude of change or providing an example from your data would demonstrate deeper insight. For Question 9.2 you identified Europe and Consulting as top performers, which aligns with your grouped summary, yet you did not discuss why these segments excel or how a business could act on this information. Adding a brief analysis of market characteristics or potential strategic actions would strengthen the response. Your answer to Question 9.3 correctly distinguishes wide versus long formats for dashboards versus calculations, but it lacks a concrete business scenario, such as using a wide format for a quarterly sales report and a long format for time-series forecasting. Providing that context would show applied understanding. In Question 9.4 you accurately describe the mechanics of `left_join` and `inner_join` and give a relevant example with customers and orders. However, you could improve by mentioning data completeness implications, such as how `left_join` preserves customers without purchases for churn analysis. Your response to Question 9.5 highlights `mutate` as valuable for creating calculated columns, which is true, but you do not explain how these new metrics (e.g., revenue per unit) drive business decisions. Connecting `mutate` to specific KPI development would make the answer more compelling.

Business Application

By flagging high-value transactions, you provide a clear basis for targeting premium customers in marketing campaigns, linking your analysis to revenue growth strategies. Your identification of Europe and Consulting as top revenue generators offers a direct input for resource allocation decisions, such as increasing sales support in those regions. The discussion of join types shows you understand how to preserve a full customer base for churn modeling while focusing on active purchasers for cross-sell opportunities.

Recommendations

Re-run your cleaning pipeline and capture before-and-after summary statistics to illustrate the effect of NA removal and outlier handling. Create a brief executive memo that translates your grouped revenue findings into actionable recommendations for regional managers. Practice reshaping a sample dataset both ways and write a one-paragraph business case for each format to solidify your understanding of when each is appropriate.

Technical Summary

- Completion: 0 out of 17 sections (0%). Calculated score: 0%.
- Completed: []
- Incomplete: [Task 1.3: Import Datasets, Task 2.2: Handle Missing Values, Task 2.3: Detect Outliers in Revenue, Task 3.3: Sort by Revenue, Task 3.4: Chain Multiple Operations, Task 4.1: Create Calculated Columns, Task 4.2: Calculate Overall Summary Statistics, Task 4.3: Regional Performance Analysis, Task 4.4: Product Category Analysis, Task 5.2: Reshape to Wide Format, Task 5.3: Reshape Back to Long Format, Task 6.2: Join Orders and Order Items, Task 7.1: Clean Text Data, Task 7.2: Parse Dates and Extract Components, Task 8.2: Calculate Business KPIs]
- This submission does not meet minimum requirements (50% completion needed for passing).
- Completion: 8/9 sections (89%). Score: 82%
- Variables found: 22/22
- Output accuracy: 69.6% (16/23 checks passed)
